

The Universal Extrapolation Space of the Shift on $C_0(\mathbb{R})$

Literature-implied full resolution of arXiv:1604.00763

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Status. `literature_implied_answer`. The conjecture is fully resolved. The decisive topological representation is published in a later paper; the short cofinality identification with the extrapolation scale is made explicit here.

Source question

Bargetz and Wegner ask in Section 4, PDF page 8 of [1], whether the universal extrapolation space for the shift semigroup on $X = C_0(\mathbb{R})$ coincides with the space $\dot{\mathcal{B}}'(\mathbb{R})$ of distributions vanishing at infinity. In their notation, the universal space is the completion of the locally convex inductive limit of the negative extrapolation spaces.

As usual in the extrapolation construction, shift the generator by a resolvent point. Write $D = d/dx$ in the distributional sense and set

$$A = I - D.$$

Both $I - D$ and $I + D$ have bounded inverses on $C_0(\mathbb{R})$ (indeed, the inverses are convolution/translation averages against one-sided exponential kernels). The shift does not change the domains $D(A^n) = C_0^n(\mathbb{R})$ or the associated universal scale up to its canonical equivalent realization.

Decisive supporting theorem

Bargetz, Nigsch, and Ortner prove on PDF page 6 of [2] the *topological* identity

$$\dot{\mathcal{B}}'(\mathbb{R}) = \varinjlim_{m \rightarrow \infty} (I - D^2)^m C_0(\mathbb{R}), \quad (1)$$

where every Banach step carries the final norm transported from $C_0(\mathbb{R})$ by $(I - D^2)^m$. They also explain why this is a Hausdorff inductive limit and why the displayed equality is topological, not merely algebraic.

Identification

Theorem 1. *Let $(T(t))$ be the translation semigroup on $C_0(\mathbb{R})$, and form its standard extrapolation scale using the invertible shifted generator $A = I - D$. Then*

$$\varinjlim_{n \rightarrow \infty} X_{-n} \cong \dot{\mathcal{B}}'(\mathbb{R})$$

topologically. In particular, the inductive limit is Hausdorff and complete, and the universal extrapolation space satisfies

$$X_{-\infty} \cong \dot{\mathcal{B}}'(\mathbb{R}).$$

Proof. First realize each abstract extrapolation step inside the distributions. By definition, X_{-n} is the completion of $X = C_0(\mathbb{R})$ for

$$\|x\|_{-n} = \|A^{-n}x\|_{\infty}.$$

If (x_k) is Cauchy for this norm, then $A^{-n}x_k$ converges in $C_0(\mathbb{R})$ to some f , while $x_k = A^n(A^{-n}x_k)$ converges distributionally to $A^n f$. Conversely, density of $D(A^n)$ in $C_0(\mathbb{R})$ gives every $A^n f$ this way. Thus, isometrically up to the harmless chosen graph-norm convention,

$$X_{-n} = E_n := A^n C_0(\mathbb{R}), \quad \|A^n f\|_{E_n} = \|f\|_{\infty}. \quad (2)$$

Put $C = I + D$ and $F_m = (I - D^2)^m C_0(\mathbb{R}) = A^m C^m C_0(\mathbb{R})$, with its transported Banach norm. We show that (E_n) and (F_m) are cofinal by continuous inclusions.

For $A^n f \in E_n$, bounded invertibility of C gives

$$A^n f = A^n C^n (C^{-n} f) \in F_n.$$

Hence $E_n \hookrightarrow F_n$ continuously. In the other direction, note that $C = 2I - A$, so

$$R := A^{-1}C = 2A^{-1} - I$$

is bounded on $C_0(\mathbb{R})$. For $A^m C^m f \in F_m$, commutativity of these constant-coefficient operators yields

$$A^m C^m f = A^{2m} R^m f \in E_{2m}.$$

Thus $F_m \hookrightarrow E_{2m}$ continuously.

These two families of inclusions induce mutually inverse continuous identity maps between the two locally convex inductive limits. Consequently

$$\operatorname{ind}_n X_{-n} = \operatorname{ind}_n E_n \cong \operatorname{ind}_m F_m \stackrel{(1)}{=} \dot{\mathcal{B}}'(\mathbb{R}).$$

The right-hand side is complete in the topology used in [2]. Therefore the completion appearing in the definition of $X_{-\infty}$ does not enlarge the limit, which proves the final assertion. $\square$

Literature status and scope

This is a full answer to the concrete $C_0(\mathbb{R})$ shift conjecture, not a solution of the source paper's broader question for arbitrary non-reflexive Banach spaces. The supporting authors cite [1], but [2] does not state that representation (1) resolves the conjecture. The relation is therefore an identified implication of published literature.

The bounded status check searched the run's registry, solution, attempt, and proof-gap indexes for the arXiv id, title, and core terms; exact-title and exact-phrase web searches; and the OpenAlex citation list for the published source article (DOI 10.1016/j.jmaa.2017.11.042). OpenAlex listed one citing work, namely [2]. No paper explicitly announcing a resolution was found. Novelty is not claimed because (1) is already a published topological theorem.

Human-review recommendation

Verify the standard realization (2) and the convention that the shift generator is replaced by the resolvent shift $I - D$. The decisive finite-step checks are the two inclusions $E_n \hookrightarrow F_n$ and $F_m \hookrightarrow E_{2m}$; both are explicit above. If those conventions match the source's universal construction, the resolution is complete.

References

- [1] C. Bargetz and S.-A. Wegner, *Pivot duality of universal interpolation and extrapolation spaces*, J. Math. Anal. Appl. 460 (2018), 321–331; arXiv:1604.00763.
- [2] C. Bargetz, E. A. Nigsch, and N. Ortner, *A simpler description of the κ -topologies on the spaces $\mathcal{D}_{L^p}$, L^p , $\mathcal{M}^1$* , Math. Nachr. 293 (2020), 1691–1706; arXiv:1711.06577.